Trolley Retrieval: A Complete Analysis with a Machine-Checked Distinguishability Theorem

Akaash Dash
akaash.dash@gmail.com

Abstract

We analyse the *Trolley Retrieval* problem (G-Research GRiddles Series A, Puzzle 4). A trolley sits on an unknown cell of a one-way infinite tape whose cell n carries the label $c(n)$, where c is a permutation of the positive integers known to the player Bob. Each turn Bob pushes the trolley one cell left or right and is told whether the label of the new cell is strictly less than the current time. Bob must eventually name the trolley's exact position; falling off the left end, or guessing wrongly, loses.

We prove the answer: **Bob has a winning strategy if and only if c is *not* eventually the identity**, i.e. $c(n) \neq n$ for infinitely many n (equivalently, c is not a finitary permutation). We give the full elementary proof of both directions. The losing direction rests on a parity lock: the threshold Bob can probe is always *even*, so an identity tail makes the pair $(2j,\ 2j+1)$ permanently indistinguishable. The winning direction rests on a *distinguishability theorem*: for a non-eventually-identity c, no pair of starting cells is indistinguishable. Its crux—previously believed to require global surjectivity bookkeeping—falls to an elementary forward-closure argument. Distinguishability is converted into a win by a single explicit, signal-independent trajectory—the *band-top lag-monotone staircase*—which we prove is safe, eventually informative, and pairwise separating; a generic reduction ("first-Yes finiteness") collapses the infinite candidate set to a finite one and turns these three properties into a winning strategy.

The losing direction, the distinguishability theorem, the candidate-collapsing reduction, the staircase construction (safety, separation, Yes-emission), and explicit winning strategies for natural "locally decodable" families are all formalised and machine-checked in Lean 4. We are scrupulous about the boundary of what is proven: the **winning direction is machine-verified modulo one true, computationally-verified combinatorial lemma**—the *anti-correlated joint-finite recurrence*—reached through the construction's combinatorial input (`band_recurrence_ge_max`) by a *joint-condition dichotomy*. For an ordered pair $a < b$ this dispatches on a single joint two-ray set; its *joint-infinite* half—which covers *all* unbounded-displacement permutations—is proven *axiom-clean* (`band_recurrence_ge_max_of_jointInfinite` via `memJ_imp_separator`), and the lone residue is the joint-finite "anti-correlated" regime. We do not claim the development is fully complete or axiom-clean; we identify precisely which results are machine-checked with no `sorry` and which single fact remains.

1 The problem

The exact statement is as follows. Let $c(1), c(2), \ldots$ be a sequence of positive integers in which every positive integer appears exactly once; equivalently c is a permutation (bijection) of $\mathbb{N}^+$. A one-way infinite sequence of cells is given: cell n (the n-th from the left, $n \geq 1$) is labelled $c(n)$.

At time $t = 0$ an invisible trolley is placed on an unknown cell, the k_0-th from the left. For each integer $t \geq 1$, in order:

1. Bob chooses a direction, left or right; the trolley moves one cell that way, to cell k_t.
2. The trolley sends Yes if $c(k_t) < t$, and No otherwise.

If ever $k_t = 0$ (the trolley falls off the left end), Bob loses immediately. At any time Bob may instead *guess* the trolley's position; a correct guess retrieves it, an incorrect guess loses.

> *Determine all permutations c for which Bob has a strategy guaranteeing safe retrieval.*

Bob knows c in full; the only hidden datum is the start cell k_0. The trolley is invisible—Bob never sees k_t directly. His sole feedback is the binary signal, and he chooses each move adaptively from the signal history.

1.1 The answer

Theorem 1.1 (Main)**.** *Bob has a strategy ensuring safe retrieval if and only if c is* not eventually the identity*; that is, if and only if $c(n) \neq n$ for infinitely many n.*

$$\boxed{\textbf{Bob wins from every start cell} \iff c(n) \neq n \text{ for infinitely many } n.}$$

We write c is *eventually identity* (EI) if there is a threshold N with $c(n) = n$ for all $n > N$. Since c is a bijection, "not EI" has several equivalent forms, all used below:

- $\{\, n : c(n) \neq n \,\}$ is infinite (c is not finitary);
- c^{-1} is not eventually identity (the condition is symmetric under $c \leftrightarrow c^{-1}$);
- c has infinitely many *upward gaps*: positions p with $c(p{+}1) - c(p) \geq 2$.

The last equivalence is worth recording, as it is exactly what fails in the EI case.

Proposition 1.2. *c has infinitely many upward gaps if and only if c is not eventually identity.*

Proof. If $c(n{+}1) \leq c(n) + 1$ for all $n > N$ (no gaps past N), then on (N, ∞) the bijection c is non-decreasing by steps of at most 1; a strictly value-preserving bijection of a tail of $\mathbb{N}^+$ that never jumps up by 2 cannot skip a value and cannot repeat one, forcing $c(n) = n$ for all $n > N$. The contrapositive is the claim. $\square$

1.2 Reformulation: the alarm-clock view and the parity lock

It is convenient to read the labels as wake-up times. Think of cell n as carrying an alarm clock set for time $c(n)$: at every time $t > c(n)$ the cell is "awake", and a trolley standing on it reports Yes; while $t \leq c(n)$ the cell is "asleep" and reports No. Because c is a bijection, exactly one cell wakes at each integer time, and every cell wakes eventually. This is just a restatement of signal = Yes $\iff c(k_t) < t$.

Let $D_t := k_t - k_0$ be Bob's net displacement at time t.

Lemma 1.3 (Parity lock)**.** *As long as Bob has not yet guessed, $D_t \equiv t \pmod 2$. Equivalently, $t - D_t$ is always even.*

Proof. $D_0 = 0$, and each move changes D by ± 1 and t by $+1$, so the parity of $t - D_t$ is invariant; it starts even. $\square$

The parity lock is the single structural fact behind the losing direction. A second, equally elementary identity exposes *why*. Let L_t be the number of *left* moves Bob has made by time t and R_t the number of right moves, so $t = L_t + R_t$ and $D_t = R_t - L_t$. Then

$$t - D_t \;=\; (L_t + R_t) - (R_t - L_t) \;=\; 2L_t. \tag{1}$$

Write $g(m) := c(m) - m$ for the displacement of the label from the diagonal. The signal at time t is

$$\begin{aligned} \textsf{Yes} &\iff c(k_0 + D_t) < t \iff g(k_0 + D_t) < t - D_t - k_0 \\ &\iff g(k_0 + D_t) < 2L_t - k_0. \end{aligned} \tag{2}$$

In other words, the walk reads the sequence g against a threshold $2L_t - k_0$ that Bob controls only through his *left-move count*.

Remark 1.4 (The whole obstruction, in one line)**.** *For the identity, $g \equiv 0$, so (2) collapses to* $\textsf{Yes} \iff k_0 < 2L_t$*. Bob can only ever compare k_0 against* even *thresholds, and $2j < 2L \iff 2j{+}1 < 2L$ for every L. Hence the start cells $2j$ and $2j{+}1$ produce* identical signals under every strategy*: the parity bit of k_0 is unrecoverable. This is the seed of the losing direction; the work below is to extend it to every eventually-identity c despite the perturbed initial segment.*

2 Losing direction: eventually identity ⇒ Bob loses

Theorem 2.1. *If c is eventually identity, Bob has no strategy that wins from every start cell.*

The proof isolates a structural "parallel-evolution" lemma and then verifies its hypothesis for two carefully chosen start cells.

2.1 Parallel evolution

Fix a strategy σ. For a start cell k let $\mathrm{pos}(k, t)$ be the position and $\mathrm{hist}(k, t)$ the signal history (most-recent first) at time t. Bob's action at any decision point is a function of hist alone.

Lemma 2.2 (Parallel evolution)**.** *Let k_1, k_2 be two start cells. Suppose that, started from k_1, Bob makes no guess before time t, and that at every time $s \le t$ the signal observed from k_1's trajectory equals the signal observed from k_2's trajectory. Then the two histories agree at time t, and the positions stay rigidly offset:*

$$\mathrm{hist}(k_1, t) = \mathrm{hist}(k_2, t), \qquad \mathrm{pos}(k_2, t) = \mathrm{pos}(k_1, t) + (k_2 - k_1).$$

Proof. Induction on t. At $t = 0$ the histories are empty and the offset is $k_2 - k_1$ by definition. For the step, equal histories make σ choose the *same* action in both worlds; since Bob has not guessed, that action is a move, applied identically to both, preserving the offset. The new signal entries are equal by the time-$(t{+}1)$ hypothesis, so the histories remain equal. □

Corollary 2.3. *If $k_1 \ne k_2$ and the two signal traces agree at* every *time under σ, then σ cannot win from both k_1 and k_2.*

Proof. Equal traces force equal histories at all times (Lemma 2.2 applied up to either guess time), hence equal first-guess times $T_1 = T_2 =: T$ and equal guesses $g_1 = g_2$. But correctness would require $g_1 = \mathrm{pos}(k_1, T)$ and $g_2 = \mathrm{pos}(k_2, T)$, while $\mathrm{pos}(k_2, T) = \mathrm{pos}(k_1, T) + (k_2 - k_1)$. These contradict $k_1 \ne k_2$. □

2.2 The indistinguishable pair for eventually-identity c

Suppose $c(n) = n$ for all $n > N$. Let M be an upper bound for the finitely many perturbed labels: $M \ge \max\{\, c(m) : 1 \le m \le N \,\}$ and $M \ge N + 1$. Choose j with $2j > M$ and set

$$k_1 = 2j, \qquad k_2 = 2j + 1.$$

Lemma 2.4 (Signal match)**.** *Fix a no-guess strategy run from $k_1 = 2j$. At every time t, every position $p \ge 1$ reachable from $2j$ in t moves (so $2j - t \le p$) that respects the parity lock ($t + p$ even) satisfies*

$$\textit{signal at } p \textit{ at time } t \;=\; \textit{signal at } p{+}1 \textit{ at time } t.$$

Proof. Two cases on the location p.

(a) $p > N$ (identity region). Both p and $p{+}1$ exceed N, so $c(p) = p$ and $c(p{+}1) = p{+}1$. The signal at p is $\textsf{Yes} \iff p < t$, and at $p{+}1$ is $\textsf{Yes} \iff p{+}1 < t$. These differ only if $t = p + 1$. But then $t + p = 2p + 1$ is odd, violating the parity hypothesis $t + p$ even. So the signals agree.

(b) $p \le N$ (perturbed region). Here the trolley has travelled from $2j$ to $p \le N$, so $t \ge 2j - p \ge 2j - N > M - N \ge \dots$; concretely $2j > M \ge N + 1$ forces $t > M$. Both $c(p)$ and $c(p{+}1)$ are at most M (using $c(N{+}1) = N{+}1 \le M$ for the boundary), hence $c(p), c(p{+}1) < t$: both signals are Yes. They agree. ☐

The two cases capture the dichotomy of Remark 1.4: in the identity tail, the parity lock makes $2j$ and $2j{+}1$ collide; in the perturbed segment, the time cost of reaching it ($t \ge 2j - N$) is so large that every cell there is already awake, so both worlds report Yes and again collide.

Proof of Theorem 2.1. Suppose σ wins from every start cell, in particular from $k_1 = 2j$ and $k_2 = 2j{+}1$ above. We claim the two signal traces agree at every time, which contradicts Corollary 2.3.

By induction on t, the trajectories from $2j$ and $2j{+}1$ evolve in lockstep with positions offset by exactly 1 and equal histories, as long as Bob has not guessed: the inductive step applies the signal match (Lemma 2.4) at $\mathrm{pos}(2j, t{+}1)$, whose hypotheses—validity $p \ge 1$ (Bob wins, so the trajectory is safe), reachability $2j - t \le p$ (a ± 1 walk), and parity (Lemma 1.3)—all hold. Hence the signal entering both histories at each step is identical, so $\mathrm{hist}(2j, t) = \mathrm{hist}(2j{+}1, t)$ and $\mathrm{pos}(2j{+}1, t) = \mathrm{pos}(2j, t) + 1$ for all t. The traces agree at every time, and Corollary 2.3 gives the contradiction. ☐

Remark 2.5 (The half-infinite tape is irrelevant here)**.** *For the chosen large $k_0 \approx 2j$, no finite-time strategy can reach cell 0, so the "falling off" rule never binds; the indistinguishability argument is unaffected by the absence of a left end.*

3 Winning direction: not eventually identity $\Rightarrow$ Bob wins

The heart of the winning direction is a converse to the losing mechanism: when c is not eventually identity, *no* pair of start cells can stay indistinguishable. We make "indistinguishable" precise, prove the theorem, and then explain how distinguishability is turned into a winning strategy—fully for two natural families, and with an honest account of the general gap.

3.1 The distinguishability theorem

Reaching displacement D from the start costs time at least $|D|$, and the parity lock forces $t \equiv D \pmod 2$. Within those constraints Bob can pad with oscillation to reach any admissible time, and (because a cell once awake stays awake) the only information at displacement D is the *set of times $t \ge D$, $t \equiv D \pmod 2$, at which the cell is awake*—equivalently, the single comparison $c(a + D) < t$. This motivates:

Definition 3.1 (Indistinguishable pair)**.** *Two start cells $a < b$ are* indistinguishable *(for c) if for every displacement $D \ge 0$ and every time $t \ge D$ with $t \equiv D \pmod 2$,*

$$c(a + D) < t \iff c(b + D) < t.$$

Quantifying over *all* reachable (D, t)—including small times that catch early wake-ups—is essential; it is exactly the freedom that lets Bob's oscillation observe a cell's wake-up time at full resolution.

Theorem 3.2 (Distinguishability)**.** *If $a < b$ are indistinguishable for c, then $b = a + 1$ and $c(n) = n$ for all $n \ge a$. Consequently, if c is not eventually identity, then* no *pair of start cells is indistinguishable.*

We prove the first sentence; the second is its contrapositive (an indistinguishable pair would exhibit an identity tail, making c EI). Throughout write $\Delta := b - a \ge 1$.

The proof has two regimes by the size of Δ. The case $\Delta = 1$ is a clean shift argument; the case $\Delta \ge 2$ is the crux. We first extract the local structure shared by both.

Lemma 3.3 (Consecutiveness)**.** *Call a displacement D* active *(for the cell a) if $c(a + D) \ge D$, and similarly for b. If $a < b$ are indistinguishable and D is active for both a and b, then $c(a + D)$ and $c(b + D)$ are consecutive integers:*

$$|\, c(a + D) - c(b + D)\,| = 1.$$

Proof. Let $X = c(a + D)$, $Y = c(b + D)$. Injectivity gives $X \neq Y$; say $X < Y$. If $Y \geq X + 2$, pick t with $X < t \leq Y$ of the parity $t \equiv D \pmod 2$ (possible because the interval $(X, Y]$ has length ≥ 2 and so contains both parities). Then $c(a + D) < t$ but $c(b + D) \not< t$, with $t \geq D$ (as D is active and $t > X \geq D$), contradicting indistinguishability at (D, t). Hence $Y = X + 1$. The case $Y < X$ is symmetric. ☐

Lemma 3.4 (Bad displacements are forward-closed)**.** *Call D* bad *(for a) if $c(a + D) < D$. If $a < b$ are indistinguishable and D is bad, then $D + \Delta$ is bad: $c(a + D + \Delta) < D + \Delta$.*

Proof. D bad means $c(a + D) < D$, i.e. the cell $a + D$ is awake at the admissible time t with $t \equiv D \pmod 2$, $D \leq t$, as soon as $t > c(a+D)$; take the least such admissible t, so $t \leq D+1$ and the signal is Yes. Indistinguishability transfers Yes to $b + D = a + (D + \Delta)$: $c(a + D + \Delta) < t \leq D + 1$, and since the value is an integer $c(a + D + \Delta) \leq D < D + \Delta$. (The point is purely that an already-saturated Yes propagates by $+\Delta$; this is where the constraint $b = a + \Delta$ enters.) ☐

Lemma 3.4 is decisive: on each residue class modulo Δ, the bad displacements form an *up-set*, so the *active* (non-bad) displacements form a finite or cofinite *prefix*—never an interleaving of good and bad. This is exactly the structural fact that earlier, more elaborate attempts (parity arguments, value-hogging bookkeeping) were trying to recover. With it, the crux is short.

Lemma 3.5 (Chain bound)**.** *If $a < b$ are indistinguishable, fix a residue r with $0 \leq r < \Delta$. For any k such that $r, r + \Delta, \dots, r + k\Delta$ are all active,*

$$c(a + r + k\Delta) \ \leq \ c(a + r) + k.$$

Proof. By Lemma 3.3, consecutive active displacements on the residue satisfy $|c(a+r+(i+1)\Delta) - c(a + r + i\Delta)| = 1$ (apply consecutiveness at the displacement $r + i\Delta$, whose $+\Delta$ shift lands at $b+r+i\Delta$). Summing k unit steps and the triangle inequality give $c(a+r+k\Delta) \leq c(a+r)+k$. ☐

Lemma 3.6 ($\Delta \geq 2$ is impossible)**.** *If $a < b$ are indistinguishable with $\Delta = b - a \geq 2$, contradiction.*

Proof. Let $M := \max_{0 \leq r < \Delta} c(a + r)$. Suppose a displacement D is active, and write $D = r + k\Delta$ with $0 \leq r < \Delta$. By forward-closure (Lemma 3.4), *all* of $r, r + \Delta, \dots, r + k\Delta$ are active (active is a prefix on each residue). Activity gives $c(a + D) \geq D = r + k\Delta$, while the chain bound gives $c(a + D) \leq c(a + r) + k \leq M + k$. Hence

$$r + k\Delta \leq M + k \quad \Longrightarrow \quad k(\Delta - 1) \leq M - r \leq M.$$

With $\Delta \geq 2$ this bounds $k \leq M$, so $D = r + k\Delta < (M{+}1)\Delta$. Thus *every* displacement $D \geq (M{+}1)\Delta$ is bad: $c(a + D) < D$ for all such D.

This forces a pigeonhole collapse. For $D \geq (M{+}1)\Delta$, the cell at position $a + D$ has label $c(a + D) < D = (a + D) - a$, i.e. $c(p) < p - a$ for all large p. The finitely many small positions have labels bounded by some constant. Choosing N large enough, c maps the segment $[1, N]$ into $[1, N - a - 1]$: large positions p land below $p - a \leq N - a - 1$, and the bounded small positions also fit once N is large. But $[1, N]$ has N elements and $[1, N - a - 1]$ has $N - a - 1 < N$, so c cannot be injective on $[1, N]$. Contradiction. ☐

Remark 3.7 (Why this is the right proof)**.** *The contradiction in Lemma 3.6 uses* no *parity argument and* no *even/odd split on Δ. The single lever is forward-closure of badness (Lemma 3.4), which turns the "active" set on each residue into a prefix; the chain bound then makes that prefix finite for $\Delta \geq 2$, and a one-line pigeonhole finishes. This is markedly simpler than—and subsumes—the earlier even/odd case analysis.*

It remains to handle $\Delta = 1$.

Lemma 3.8 ($\Delta = 1$ forces an identity tail)**.** *If $a < b = a + 1$ are indistinguishable, then $c(n) = n$ for all $n \geq a$.*

Proof. For $\Delta = 1$, $b + D = a + (D + 1)$, so consecutiveness (Lemma 3.3) on active displacements reads $|c(a + D + 1) - c(a + D)| = 1$: the labels along $a, a{+}1, a{+}2, \dots$ form a self-avoiding ± 1 walk. Let $\sigma(0) = \operatorname{sgn}\big(c(a{+}1) - c(a)\big)$ be the sign of the first step. If $\sigma(0) = -1$ (descending start), the walk $c(a), c(a){-}1, c(a){-}2, \dots$ would eventually drop below 1, impossible for labels in

$\mathbb{N}^+$; so $\sigma(0) = +1$. A self-avoiding ± 1 walk that starts ascending must ascend forever (any later descent would revisit a value, breaking injectivity), giving $c(n+1) = c(n) + 1$ for all $n \geq a$, i.e. $c(n) = n + C$ on $[a, \infty)$ with $C = c(a) - a$. If $C > 0$, the values $\{1, \ldots, C\}$ are absent from $c([a, \infty))$ and must be supplied by the finite set $c([1, a-1])$, which has only $a-1$ elements; bijectivity of c on $\mathbb{N}^+$ then forces $C = 0$ (and $C < 0$ was already excluded). Hence $c(n) = n$ for $n \geq a$. (In Lean this sign analysis is the pair `indist_Delta1_pos` / `indist_Delta1_neg`.) ☐

Proof of Theorem 3.2. Let $a < b$ be indistinguishable, $\Delta = b - a$. By Lemma 3.6, $\Delta \geq 2$ is impossible, so $\Delta = 1$, i.e. $b = a + 1$; then Lemma 3.8 gives $c(n) = n$ for $n \geq a$, so c is eventually identity. Contrapositively, if c is not eventually identity then no pair is indistinguishable. ☐

3.2 From distinguishability to a strategy: the obstruction

Theorem 3.2 says that for a non-EI c every pair of cells admits a separating observation. Converting this into a single winning strategy is the substantial part of the problem, and naive attempts fail in instructive ways; we record the obstruction now and resolve it in Section 4.

The lag is monotone—and that is the whole difficulty. Recall from (1) that the *lag* $\ell_t := t - D_t = 2L_t$ equals twice Bob's left-move count. Since each move changes D by ± 1 and t by $+1$, we have $\ell_{t+1} - \ell_t \in \{0, 2\}$: the lag is **monotone nondecreasing** and never returns to a smaller value. A separating observation for a pair (a, b) occurs at a specific displacement D and a specific *slack* $s = t - D = \ell_t$; by the alarm-clock identity (2) it is the event

$$\big[\, c(a + D) < D + s \,\big] \;\neq\; \big[\, c(b + D) < D + s \,\big].$$

Because the lag only increases, a separator at slack s is *use-it-or-lose-it*: once Bob's lag has passed s, he can never again present that pair with that slack. He must therefore visit separators in nondecreasing slack order.

The tension. This pits two requirements against each other:

- *Catching a low-slack discriminator* (an *early* wake-up, where t is close to D) requires Bob to be *riding the diagonal*: large displacement D, small slack. Once missed it is gone.
- *Bounding the start cell* k_0 requires Bob to be *near the origin*: small displacement, large slack, so that the threshold $\ell_t - k_0$ in (2) grows and probes a window of cells around k_0.

At any single instant Bob is at one (D, ℓ); he cannot be both near the origin and far down the diagonal. Reconciling the two—monitoring early discriminators while still bounding k_0—is the genuine combinatorial heart of the winning direction. Section 5 records three simulation-verified negative results showing that no shortcut (no fixed universal trajectory, no two-phase "bracket then navigate", no free-standing finite-belief induction) reconciles them; the resolution in Section 4 is a c-tailored staircase that interleaves the two regimes.

We first give complete strategies for two natural families where the tension does not arise because every discriminator sits at a *bounded* displacement.

3.3 Two complete winning families (locally decodable)

When the candidate pair is always separated by a wake-up that occurs at a *bounded* displacement, Bob can resolve it by a fixed-amplitude oscillation that monitors a small window from time 0—catching even the early wake-ups that a navigate-then-observe approach would arrive too late to see. Two regimes are formalised.

Definition 3.9 (Locally $K{=}2$ decodable)**.** *c is* locally decodable *if the map $k_0 \mapsto \big($first even* Yes *time of $c(k_0)$, first odd* Yes *time of $c(k_0{+}1)\big)$ is injective; equivalently the bucketed pair $(c(k_0), c(k_0{+}1))$ determines k_0.*

Theorem 3.10 ($K{=}2$ winning theorem)**.** *If c is locally decodable, Bob wins. The strategy oscillates $D \in \{0, 1\}$, reads off $c(k_0)$ and $c(k_0{+}1)$ from the first even and first odd* Yes*, decodes k_0, and guesses its current position—never leaving $\{k_0, k_0{+}1\}$, so safety is automatic.*

Example 3.11 (Adjacent transpositions)**.** *Let c swap $2i{-}1 \leftrightarrow 2i$ for every i (so $c(n) = n + 1$ if n is odd, $c(n) = n - 1$ if n is even). Here $g(n) = c(n) - n \in \{+1, -1\}$ is bounded, yet c is not eventually identity; Bob wins by local decoding. This already shows the boundary is non-EI, not unbounded growth of g.*

Example 3.12 (Block 3-cycles on consecutive triples)**.** *Let c cyclically rotate each triple $\{3i{-}2, 3i{-}1, 3i\}$, e.g. $3i{-}2 \mapsto 3i{-}1 \mapsto 3i \mapsto 3i{-}2$. This is locally decodable (the bucketed pair of adjacent labels recovers k_0), and Bob wins.*

Some non-EI permutations need a third probed label. The block 3-cycle written "forwards" is the canonical example.

Definition 3.13 (Locally $K{=}3$ decodable)**.** *c is* locally 3-decodable *if the triple of first-*Yes *times at displacements $0, 1, 2$—equivalently the bucketed triple $(c(k_0), c(k_0{+}1), c(k_0{+}2))$—determines k_0.*

Theorem 3.14 ($K{=}3$ winning theorem)**.** *If c is locally 3-decodable, Bob wins, via a period-4 triangle-wave sweep of displacements $0, 1, 2, 1, 0, 1, 2, \ldots$ with three detectors (first* Yes *at displacements 0, 1, 2). The displacement stays in $[0, 2]$, so safety is automatic.*

Example 3.15 (Forward block 3-cycle)**.** *Let $c(m) = m{+}1$ unless $3 \mid m$, in which case $c(m) = m{-}2$ (so each block $\{3b{+}1, 3b{+}2, 3b{+}3\}$ rotates forwards: $c(1) = 2$, $c(2) = 3$, $c(3) = 1$). Then $c(m) \le m + 1$ always, so* no *rightward probe $[c(k_0{+}D) < T_1{+}D]$ ever separates the Phase-1 candidates—a navigate-only strategy fails. But cell 3 wakes early, at time 2 (since $c(3) = 1$), which splits $k_0 = 1$ from $k_0 = 2$ at displacement 1: at ($D{=}1$, $t{=}3$), the world $k_0 = 2$ stands on the already-awake cell 3 and reports* Yes*, while $k_0 = 1$ stands on cell 2, still asleep ($c(2) = 3 \not< 3$), and reports* No*. The $K{=}3$ sweep, monitoring small displacements from time 0, catches this early wake-up. So c is locally 3-decodable, and Bob wins.*

Example 3.15 is instructive: it is a concrete non-EI Bob-win whose resolution *requires* catching an early wake-up at a small displacement, which is precisely the regime the local decoders are built for—and precisely what naive "oscillate then navigate right" strategies miss.

Remark 3.16 (The fixed-K architecture cannot cover all non-EI c)**.** *The locally-K-decodable classes (any fixed amplitude K, formalised in Lean for arbitrary $K \in \mathbb{N}^+$) are genuinely incomplete. The* block-5 cycle*—rotate each $\{5b{+}1, \ldots, 5b{+}5\}$ forward—is non-EI, yet for* every *K some pair of start cells produces an identical $K{+}1$-tuple of first-*Yes *times (checked exhaustively for $K = 1, \ldots, 50$): the colliding pair $(4, 5)$ has $c(4..9) = (5, 1, 7, 8, 9, 10)$ and $c(5..10) = (1, 7, 8, 9, 10, 6)$, which a coarse amplitude-K sweep cannot tell apart even though Theorem 3.2 guarantees them distinguishable (here at $D{=}0$, $t{=}2$: $c(4) = 5 \ge 2$ but $c(5) = 1 < 2$). A single fixed-amplitude trajectory cannot finely monitor all displacements at once. The next section gives a c-tailored trajectory that does cover every non-EI c.*

3.4 Worked examples table

Permutation c	**EI?**	**Bob**	**Reason**
Identity $c(n) = n$	yes	**loses**	$(2j,\ 2j{+}1)$ collide: only even thresholds probe-able (Rem. 1.4).
Finitary (any finite-support perm.)	yes	**loses**	EI with $N =$ support bound; same collision for large k_0.
Adjacent transpositions	no	**wins**	locally 2-decodable (Ex. 3.11); bounded g.
Block 3-cycle (rotated triples)	no	**wins**	locally 2- or 3-decodable (Ex. 3.12, 3.15).
Block 5-cycle (rotated quintuples)	no	**wins**	not EI; needs the staircase—no fixed-K decoder suffices (Rem. 3.16).
Any non-EI c	no	**wins**	band-top staircase separates every pair (§4, Thm. 3.2).

Table 1: Outcomes by family. Eventually-identity $\iff$ Bob loses.

4 The winning strategy for every non-EI c

We now give the construction that wins for an arbitrary non-eventually-identity c. It has two layers. First, a generic *reduction*: any single signal-independent trajectory that is safe, eventually informative, and pairwise separating already constitutes a winning strategy. Second, an explicit such trajectory: the *band-top lag-monotone staircase*. The reduction is fully proven and axiom-clean; the staircase's three properties are proven modulo one isolated combinatorial recurrence lemma (the anti-correlated joint-finite regime), reached by a joint-condition dichotomy whose joint-infinite half—covering all unbounded-displacement c—is itself proven axiom-clean (Sections 6, 7).

4.1 The reduction: separation + safety + Yes $\Rightarrow$ Bob wins

Fix a deterministic move sequence $m \colon \mathbb{N} \to \{\mathrm{L}, \mathrm{R}\}$ and let Bob play it *ignoring the signals for his moves*, using the signals only to decide *when to stop and guess*. Write $\mathrm{pos}_m(k_0, t)$ for the trajectory's position from start k_0. Call the trajectory:

- *safe* if $\mathrm{pos}_m(k_0, t) \geq 1$ for all k_0, t (never falls off the left);
- *Yes-emitting* if from every k_0 some time emits Yes;
- *separating* if for every $a \neq b$ the signal histories from a and from b eventually differ.

Theorem 4.1 (Reduction; `localizes_BobWins`, `ofMoves_localizes`). *If a fixed trajectory m is safe, Yes-emitting, and separating, then Bob wins from every start cell.*

The nontrivial point is that separation is an *infinite* family of conditions (one per wrong candidate), whereas a win requires a *single finite* stop time T that has ruled out *all* wrong candidates at once. The bridge is first-Yes finiteness.

Lemma 4.2 (First-Yes finiteness; `yesSet_finite`). *For any time s and displacement-offset δ, the set of start cells k whose signal at time s (after the shift δ) is Yes is* finite.

Proof. A Yes at (s, δ) from start k means $c(k + \delta) < s$. As c is a bijection, only the finitely many values $\{1, \ldots, s-1\}$ lie below the threshold, so only finitely many cells $k+\delta$ (hence finitely many k) can emit Yes at that moment. $\square$

Proof of Theorem 4.1. Fix the true start k_0. By Yes-emission there is a first time s_0 at which k_0's trajectory reads Yes. Any candidate k *not yet distinguished* from k_0 by time s_0 must read the *same* signal, hence also Yes, at s_0; by Lemma 4.2 there are only finitely many such k. Each of these finitely many candidates is separated from k_0 at *some* finite time (separation), so the maximum T of those finitely many separation times is finite, and by time T *every* wrong candidate has produced a differing signal. Bob plays m until T; the histories now pin k_0 uniquely. Knowing k_0 and his own (signal-independent) displacement D_T, Bob computes the current cell $k_0 + D_T$ and guesses it. Safety guarantees every visited position was a valid cell, so the guess is timely and correct. $\square$

In Lean, the "stop and guess" wrapper is the belief-state strategy `beliefStrat`: it replays m's moves until the set of signal-consistent candidates is a singleton, then guesses. The match between the wrapped and bare trajectories up to the first resolved time, and the correctness of the guess, are discharged in `localizes_BobWins` (axiom-clean). What remains is to exhibit one fixed trajectory with the three properties.

4.2 The band-top lag-monotone staircase

Recall the lag $\ell_t = t - D_t = 2L_t$ is monotone (§3.2), and a separator for (a, b) at displacement D lives at a definite even *slack* s, where the trajectory must have lag exactly s. The construction therefore organises the work by slack.

Recurring slack. For distinct a, b, call an even slack s *recurring* if slack-s separators for (a, b) occur at arbitrarily large displacements: $\forall D_0 \, \exists D \geq D_0$ with a slack-s separator at D. The construction needs, for each pair, *one* recurring slack $s(a, b)$, and it needs $s(a, b) \geq \max(a, b)$ (explained below). The existence of such a slack is the sole combinatorial input (Theorem 6.1, discharged in Lean by the joint-condition dichotomy down to a single residual lemma, §6).

The staircase. Enumerate the ordered distinct pairs (a, b), $a < b$, by the key $\big(s(a, b), \max(a, b), \min(a, b)\big)$ in increasing order. Because $s(a, b) \geq \max(a, b)$, only finitely many pairs have any given key value (at most $\binom{S}{2}$ pairs have $s \leq S$, since $\max(a, b) \leq S$), so the keys form a well-order of order type ω: every pair has only finitely many predecessors. This ω-ordering is exactly why the lower bound $s(a, b) \geq \max(a, b)$ is load-bearing—without it, infinitely many pairs could share one slack level and the enumeration would not have type ω.

Process the pairs in this order. The state is the current (D, ℓ) with $t = D + \ell$, starting at $(0, 0)$. Two primitives keep $D \geq 0$: **R** appends a right move ($D+=1$, lag fixed), and **L** appends a left move (*only when* $D \geq 1$; $D-=1$, $\ell+=2$). For pair (a, b) with target slack $s := s(a, b)$:

1. **Raise** the lag from its current (smaller, by the key order) value to s, oscillating $D \in \{0, 1\}$ via R,L pairs—each pair bumps ℓ by 2 while returning D to its prior value.
2. **Ride** right at fixed lag s until the displacement lands on a slack-s separator for (a, b); one exists at unbounded displacement by recurrence, so the ride terminates.
3. **Tail**: ride right until the displacement exceeds the phase index, forcing displacements to grow without bound across phases.

In the Lean formalisation this is a small state machine (`trajSt`, `stairMoves`) with sub-phases `raiseR/raiseL/ride`; the identity $t = D + \ell$ with even lag $\ell = 2k$ ($+2$ while riding) is the invariant `time_eq_D_add_lag`.

Lemma 4.3 (Safety; `stair_safe`)**.** *The staircase never falls off: from any $k_0 \geq 1$, position $= k_0 + D_t \geq 1$.*

Proof. $D_t \geq 0$ always: **R** raises D, and **L** is emitted only at $D \geq 1$ (Lean `raiseL_D_pos`), so it lands at $D \geq 0$. Hence $\mathrm{pos} = k_0 + D_t \geq 1$. ☐

Lemma 4.4 (Separation; `stair_separates`)**.** *For every $a \neq b$, the staircase's histories from a and b eventually differ.*

Proof. The pair is processed at a finite phase (its predecessors in the key-order are finite). At that phase the ride at lag $s(a, b)$ lands on a slack-$s(a, b)$ separator, where by definition the signals at the common displacement differ. (Lean: `separates_OPair`, via the slack-bucket structure `bucket/keyOf`.) ☐

Lemma 4.5 (Yes-emission; `stair_yes`)**.** *From every start k_0 the staircase eventually reads a* Yes.

Proof. By (2) a Yes at displacement D is $c(k_0 + D) < D + \ell$, i.e. $g(k_0 + D) < \ell - k_0$. The lag $\ell \to \infty$ (the targets $s(a, b) \geq \max(a, b) \to \infty$), and every displacement lies in some phase's ride window (`ride_window_cover`). Pick a phase of lag $\ell > k_0$ and a *weak descent* of k_0—a displacement D with $c(k_0 + D) \leq k_0 + D$, of which there are infinitely many for any permutation (`weak_descents_infinite`, axiom-clean). Then $c(k_0+D) \leq k_0+D < (k_0+D)+(\ell-k_0) = D+\ell$, a Yes. ☐

Remark 4.6 (Why a fixed trajectory suffices after all)**.** *Earlier analysis (Section 5) suggested the winner had to be a genuinely adaptive belief-state machine. The resolution is that the* moves *can be fixed (signal-independent) provided the trajectory is tailored to c; Bob uses the signals only to choose his stop time. The staircase's move schedule depends on c (through the recurring slacks $s(a, b)$ and the ride lengths), but it is a single predetermined sequence, which is exactly what the reduction (Theorem 4.1) consumes.*

Combining the three lemmas with Theorem 4.1 proves the winning direction: every non-EI c is a Bob-win (`notEI_BobWins`), modulo the existence of the recurring slacks (Theorem 6.1), whose joint-infinite half is already axiom-clean and to which we now turn.

5 Prior negative results: why the strategy is as it is

Three simulation-verified negative results explain why simpler strategies do not work, and hence why the staircase architecture is needed. All were established by exact-dynamics simulation respecting the parity lock.

1. **No fixed *universal* trajectory.** A single *c-independent* move schedule cannot work for all non-EI c. Several natural candidates (fixed-K triangle wave, growing triangle, slow "dwelling" sweep, sawtooth "right a / left 1") were tested against random period-B permutations; even the best sawtooth—which separates all $k_0 \leq 40$ for block-5, reverse-block-5, block-3, and adjacent transpositions—still *collides* on 71 of 1247 random periodic non-EI c. The move schedule must read structure from c. (This is why the staircase's slacks $s(a, b)$ are c-dependent.)

2. **Two-phase "bracket then navigate" fails.** The natural plan—oscillate at $D = 0$ to pin $c(k_0) \in \{\tau{-}2, \tau{-}1\}$ (the first-Yes time τ is even, so the two survivors have *consecutive* c-values), then navigate to a discriminator—is doomed because the surviving pair's discriminators may all have *already expired*. For block-5 the pair $(1, 2)$ separates only at slack $t - D \leq 2$ (e.g. $D{=}3, t{=}3$), all of which lie in the past once the bracket completes at $\tau = 4$; Bob cannot navigate backward in time. This is the low-slack-vs-bounding tension of §3.2 made concrete.

3. **"Finite belief $\Rightarrow$ winnable" is false.** Winnability is *not* a function of how many candidates remain. A two-element belief reached at a late state whose only discriminators lie in the past is genuinely lost. Winnability is a property of *reachable, non-stranding* states—which is precisely what the monotone-lag staircase guarantees by visiting separators in nondecreasing slack order, never overshooting a pair's slack before separating it.

The common root cause is the lag-monotonicity tension (§3.2): catching low-slack discriminators wants $D \approx t$ (riding the diagonal); bounding k_0 wants D small (near the origin). They conflict at every instant, and only a slack-ordered interleaving—the staircase—reconciles them.

6 The residual lemmas

The entire winning construction rests on a single combinatorial input: that each pair has a recurring slack $s(a, b) \geq \max(a, b)$.

Theorem 6.1 (Band recurrence at a max-bounded slack; `band_recurrence_ge_max`). *For non-eventually-identity c and any distinct a, b, there is an* even *slack $s \geq \max(a, b)$ whose slack-s separators recur at arbitrarily large displacement: $\forall D_0\, \exists D \geq D_0$ such that $[\, c(a{+}D) < D{+}s \,] \neq [\, c(b{+}D) < D{+}s \,]$.*

This is sharper than what the axiom-clean machinery already delivers. `separators_unbounded` (proven, axiom-clean) gives separators at unbounded displacement but at *possibly unbounded slack*; the slack can drift to ∞. Theorem 6.1 additionally *pins the slack* into a fixed finite window $[\max(a, b),\, \max(a, b) + O(1)]$, and it is this fixed-slack recurrence that the staircase's per-pair ride needs (it rides at one lag and must meet a separator there).

The joint-condition dichotomy. In Lean, Theorem 6.1 is *not* discharged by a single all-slacks argument (an earlier route through such a lemma is recorded as a defeated approach in §6.1). Instead, for an ordered pair $a < b$, set $M := b$, $p := M \bmod 2$, and fix the *even* slack $s := M + p$ (so $s = M$ when M is even, $s = M + 1$ when M is odd; in all cases $s \geq \max(a, b)$). Define the **joint set**

$$J \;:=\; \big\{\, D \;:\; c(a{+}D) \leq a{+}D \;\text{ and }\; c(b{+}D) \geq b{+}D + p \,\big\},$$

the displacements at which a *weak descent of the lower cell* a coincides with a *parity-corrected ascent of the upper cell* b. The dichotomy splits on whether J recurs at unbounded displacement:

- **Joint-infinite** (J unbounded). *Proven axiom-clean* (`band_recurrence_ge_max_of_jointInfinite` via `memJ_imp_separator`). At every $D \in J$ the fixed slack $s = M + p$ separates the pair: the weak descent of a puts the window's lower edge $\min\big(c(a{+}D), c(b{+}D)\big) \leq c(a{+}D) \leq a{+}D < M{+}D \leq D{+}s$ (using $a < b = M$), while the parity-corrected ascent of b puts the upper edge $\max\big(c(a{+}D), c(b{+}D)\big) \geq c(b{+}D) \geq b{+}D{+}p = D{+}s$. Hence $D{+}s \in (\min, \max]$ is a slack-s separator (`separatorAtSlack_iff_window`), and the recurrence transports directly. This case **covers the entire unbounded-displacement regime**: every unbounded c is joint-infinite on every pair (verified), so the former separate unbounded-case residue is *removed*.
- **Joint-finite** (J bounded). The lone residue, `band_recurrence_ge_max_of_jointFinite`. This is the single "anti-correlated" regime in which weak-descents of a and parity-corrected

ascents of b never co-occur cofinitely (the canonical witness is the *parity-shift* permutation $c(2k){=}2k{+}2$, $c(2k{+}1){=}2k{-}1$, whose J is empty for that ordered pair). It is empirically bounded-displacement, true, and computationally verified (§below), but unformalised.

The joint-finite residue is *true*—verified over $\sim 10^4$–10^7 cases with zero counterexamples (§below)—but not yet formally proven; it is the **only** `sorry` in the development.

Why it is true. `weak_descents_infinite` (proven, axiom-clean) gives infinitely many D with $c(a{+}D) \le a{+}D$, which forces the *lower* cell's contribution into the slack window $[\max(a,b), \max(a,b){+}O(1)]$; `weak_ascents_infinite` (axiom-clean) does the same for the upper edge via the ascent of b. The joint-infinite case is precisely the statement that these two infinite sets—$\{D : c(a{+}D) \le a{+}D\}$ and $\{D : c(b{+}D) \ge b{+}D{+}p\}$—meet infinitely often, and there it is *proven*. Only when they fail to align cofinitely (the joint-finite regime) does the band recurrence still hold by a global counting argument that no single bijectivity engine discharges. Theorem 6.1 as a whole has been **decisively verified computationally**: over 2300 (family, pair) combinations—block-B cycles, reverse blocks, the adjacent-transposition involution, sparse swap-at-powers, the parity-shift $c(2k) = 2k{+}2$, $c(2k{+}1) = 2k{-}1$, unbounded-growth growing-block and growing-reverse-block families, random period-B permutations, random bijections on a $6 \cdot 10^4$ prefix, and large-displacement involutions—with *zero* counterexamples, and the recurring-slack excess $s - \max(a, b)$ is uniformly ≤ 2 (the canonical witness $s = M + (M \bmod 2)$ works for almost every pair; only the parity-shift family needs $\max{+}2$). Crucially, the joint discriminator confirmed that *no* unbounded family is joint-finite, so the axiom-clean joint-infinite case absorbs the whole unbounded regime.

6.1 Approaches considered for the residual lemma

The lemma is not for want of trying. We first record three natural proof routes for the original *all-slacks* formulation, each ruled out by an explicit non-EI permutation that defeats it; together they explain why that formulation is genuinely hard rather than merely unfinished. A literature search located only analogues—the parity-sweep of Hunters-and-Rabbit pursuit games and a König/min-cut argument—none of which applies off the shelf. We then describe the convergent-permutation classification we tested the split against, and finally the joint-condition dichotomy that supersedes the all-slacks route and is what the Lean development now uses.

1. **Slack-localised Lemma A (information deficit).** The natural hope is to re-run the distinguishability engine (Lemma 3.3) at a single fixed slack. But the contrapositive hypothesis "no slack-s separator for all $D \ge D_0$" supplies signal *agreement at exactly one time $t = D + s$ per displacement*—a single bit per D. The engine that pins $|c(a + D) - c(b + D)| = 1$ reads the signal at *three* distinct times per displacement; one bit per D simply does not reproduce the consecutive-values conclusion. Moreover the universal quantifier over slacks is load-bearing: the *parity-shift* permutation $c(2k) = 2k{+}2$, $c(2k{+}1) = 2k{-}1$ agrees at a single fixed slack forever yet is non-EI, so no *one* slack can be hard-wired. (This is why the existing axiom-clean `IndistFrom` machinery, which floors the displacement rather than the slack, does not supply the lemma.)
2. **Band-collapse with cut counting.** A second route models the agreement constraints as a flow/cut on the bipartite incidence of cells against awake-times and tries to close the band by a min-cut argument. The *same* parity-shift permutation $c(2k) = 2k{+}2$, $c(2k{+}1) = 2k{-}1$ defeats it: it is non-EI yet has *balanced, residue-constant* displacement $g(n) = c(n) - n \in \{+2, -2\}$, so the cut never closes—the collapse argument certifies agreement that the permutation does not in fact exhibit. A displacement-floored closure cannot see the parity structure that actually separates the pair.
3. **Positive-density averaging.** A third route would derive the recurrence from a positive lower density of separators (so that two positive-density sets must intersect). This fails because the recurrence is intrinsically *sparse*: the *swap-at-powers* permutation (which transposes only cells indexed by powers of 2) is non-EI yet has only $O(\log N)$ separators among the first N displacements, a density tending to 0. The two infinite displacement-sets must therefore be coordinated *exactly*, not merely on average.

The honest characterisation of the all-slacks formulation is that it is a *cofinitely-sparse coordination of two infinite displacement-sets* ($\{D :$ a separator exists$\}$ and $\{D : c(a + D) \le a + D\}$, which must meet at a common slack-bounded displacement infinitely often). It survives only in the cofinite-D

limit via a global counting argument. This is the common wall the three routes above hit: the per-slack displacement floor does not uniformise across slacks.

Convergent-permutation classes (C/D), tested and discarded. A natural hope is to split the lemma along the classical convergent/divergent dichotomy for permutations of $\mathbb{N}$ (Agnew, Bourbaki, Kronrod; surveyed by Wituła). For a permutation p, let $t(p,n)$ count the maximal consecutive runs in $p(\{1,\ldots,n\})$; p is *convergent* (class C) when $\limsup_n t(p,n) < \infty$ and *divergent* (class D) when $t(p,n) \to \infty$, and Wituła's Lemma 7.3 supplies a block-count identity relating the "ascent" and "descent" index sets $U(p), V(p)$. We classified every documented witness and found the split *does not align with the difficulty boundary*: the parity-shift, swap-at-powers, growing-reverse-block, and all block/random-periodic/adjacent-involution witnesses are *all class C* (bounded block count), so the five failed routes all live in class C. A purpose-built genuine class-D non-EI permutation has $U(p) \cup V(p)$ covering a 0.995 density of cells, making the block-count machinery vacuous on the per-pair question. The finer Johnston subgroup split within class C (intervals move boundedly) likewise fails to isolate the hard cases (the hardest wall, parity-shift, is inside it while a non-member is handled by the same window engine). The classification supplied clean *vocabulary* to test the split cleanly, and the verdict was negative.

The joint-condition dichotomy (the route now used). The route that does make progress is the joint-condition split of §6: for an ordered pair $a < b$, dispatch on whether the joint set $J = \{D : c(a+D) \leq a+D \wedge c(b+D) \geq b+D+p\}$ (with $p = b \bmod 2$) recurs at unbounded displacement. Its breakthrough is the *joint-infinite* half, `band_recurrence_ge_max_of_jointInfinite` (*proven axiom-clean* via the window arithmetic `memJ_imp_separator`): whenever the weak-descent ray of a meets the parity-corrected ascent ray of b, the *fixed* even slack $s = b+(b \bmod 2) \geq \max(a,b)$ separates the pair, so the recurrence follows outright. This is exactly the uniformity the all-slacks wall lacked—a single slack, pinned by the two bijectivity engines (`weak_descents_infinite`, `weak_ascents_infinite`), with no per-slack floor to uniformise. Decisively, *every* unbounded-displacement c is joint-infinite (verified), so the joint-infinite case absorbs the entire unbounded regime—the swap-at-power log-sparsity of route 3 and the former unbounded residue both disappear. The only surviving case is the *joint-finite* "anti-correlated" regime, `band_recurrence_ge_max_of_jointFinite`: the parity-shift permutation that defeated routes 1 and 2 above is its canonical witness (J empty for the relevant pair), and there the two rays never align cofinitely.

The sole surviving residue is therefore the *joint-finite anti-correlated recurrence*; the joint-infinite half—covering all unbounded-displacement permutations—is proven axiom-clean. The residue is the genuine combinatorial heart of the winning direction—the one piece not yet given a Lean proof.

7 Machine-verification status

The argument is formalised in Lean 4 (with Mathlib) in the development `griddles-p4-lean`; `lake build` succeeds (3334 jobs). We report the status precisely. A substantial core—the losing direction, the distinguishability theorem, the *entire candidate-collapsing reduction*, and the full K-decodable family—is machine-checked and *axiom-clean*, and the winning direction is fully formalised (the staircase's safety, separation, and Yes-emission lemmas are all proved) **modulo one combinatorial lemma** reached from Theorem 6.1 (`band_recurrence_ge_max`) by the joint-condition dichotomy of §6: the *joint-infinite* case (`band_recurrence_ge_max_of_jointInfinite`), which covers all unbounded-displacement c, is itself *proven axiom-clean*, and the lone residue is the *joint-finite* anti-correlated case `band_recurrence_ge_max_of_jointFinite`. This is the *only* `sorry` in the entire development. We do *not* claim the main theorem is fully proven or axiom-clean.

For each item, "axiom-clean" means the Lean `#print axioms` audit lists only the three standard foundational axioms `propext`, `Classical.choice`, `Quot.sound`, with *no* `sorryAx`. Results carrying `sorryAx` do so *solely* through the single joint-finite residue of Theorem 6.1.

What is machine-checked, no `sorry`. The losing direction; the distinguishability theorem (Theorem 3.2) including its $\Delta \geq 2$ crux by the elementary forward-closure argument; the separator inputs (`separators_unbounded`, `weak_descents_infinite`, `weak_ascents_infinite`); the *joint-infinite* case of the band recurrence (`band_recurrence_ge_max_of_jointInfinite`, via the window arithmetic `memJ_imp_separator`), which covers *all* unbounded-displacement c; the candidate-

Component	**Status**	**Lean name**
Losing direction (Thm. 2.1)	axiom-clean	`EventuallyIdentity_loses`
Distinguishability (Thm. 3.2)	axiom-clean	`Indistinguishable_implies_eventually_identity`; `not_eventually_identity_implies_distinguishable`
Separators at unbounded displacement	axiom-clean	`separators_unbounded`
Infinitely many weak descents / ascents	axiom-clean	`weak_descents_infinite`, `weak_ascents_infinite`
Joint set fixed-slack separator	axiom-clean	`memJ_imp_separator`
Joint-infinite case (covers unbounded)	axiom-clean	`band_recurrence_ge_max_of_jointInfinite`
First-Yes finiteness (Lem. 4.2)	axiom-clean	`yesSet_finite`
Reduction (Thm. 4.1)	axiom-clean	`localizes_BobWins`, `ofMoves_localizes`
Staircase safety / separation / Yes	via Thm. 6.1	`Staircase.stair_safe`, `Staircase.stair_separates`, `Staircase.stair_yes`
Separating fixed trajectory exists	via Thm. 6.1	`separating_trajectory_exists`
$K \in \mathbb{N}^+$ locally-decodable winning classes	axiom-clean	`LocallyDecodable_BobWins` (K=2), `LocallyK3Decodable_BobWins` (K=3), `LocallyKDecodable_BobWins` (any K)
Capstone (decodable consistency)	axiom-clean	`locallyDecodable_main`
Residue: joint-finite (anti-correlated)	**the lone `sorry`**	`band_recurrence_ge_max_of_jointFinite`
Band recurrence (Thm. 6.1)	via the residue	`band_recurrence_ge_max` (dispatches on joint set J)
Winning direction (§4)	via Thm. 6.1	`notEI_BobWins`
Main *iff* (Thm. 1.1)	via Thm. 6.1	`main`

Table 2: Formalisation status. "axiom-clean" = no `sorryAx` (only the three standard axioms). "via Thm. 6.1" / "via the residue" = the result's only `sorryAx` flows through the single joint-finite residual lemma. The losing direction, the distinguishability theorem, the candidate-collapsing reduction, the proven-axiom-clean separator inputs, the axiom-clean joint-infinite case (covering all unbounded-displacement c), and the entire $K \in \mathbb{N}^+$ decodable family are machine-checked with no `sorry`.

collapsing reduction (Theorem 4.1, including first-Yes finiteness); and the complete $K \in \mathbb{N}^+$ locally-decodable winning family with its concrete instances (`adjPerm`, `cyc3Perm`, `blkPerm`). All of these pass the `#print axioms` audit with no `sorryAx`. (The earlier bounded-displacement scaffolding—Step A `separatorAtSlack_slack_le_of_bddDisp` and the finite pigeonhole `fixedSlack_ge_of_window_recurrence`—remains in the development, axiom-clean, but is superseded by the joint dichotomy and no longer feeds `main`.)

What the one `sorry` is. The development contains **exactly one** `sorry` in total: the joint-finite residue of Theorem 6.1 reached by the joint-condition dichotomy, `band_recurrence_ge_max_of_jointFinite` (the anti-correlated regime in which weak-descents of the lower cell and parity-corrected ascents of the upper cell never co-occur cofinitely). Its companion *joint-infinite* case (`band_recurrence_ge_max_of_jointInfinite`), covering the entire unbounded-displacement regime, is proven axiom-clean. Every result that carries `sorryAx`—`band_recurrence_ge_max`, `separating_trajectory_exists`, the three staircase lemmas `Staircase.stair_safe/stair_separates/stair_yes`, `notEI_BobWins`, and `main`—inherits it from this single lemma alone, and from nothing else. There are no orphan `sorry`s: an earlier all-slacks route (through the now-deleted lemma `noSeparator_allSlack_imp_eventuallyIdentity`) and the prior two-residue displacement dichotomy (its bounded case `band_recurrence_finite_window_of_bddDisp` and unbounded case `band_recurrence_ge_max_of_unbddDisp`, both now deleted) were retired as dead code; their defeated approaches are archived in the project notes. The band-top staircase now consumes `band_recurrence_ge_max` directly, leaving the project with this one `sorry` and no others, as the `#print axioms main` audit (`[propext, sorryAx, Classical.choice, Quot.sound]`) confirms.

Summary—honest boundary. The *answer* (Theorem 1.1) is settled, and the *architecture* of both directions is fully formalised: the losing direction and distinguishability theorem are machine-checked and axiom-clean, and the winning direction is reduced—through an axiom-clean reduction and a fully formalised staircase construction whose safety, separation, and Yes-emission lemmas are all proved—to one explicit, precisely stated, computationally-verified ($\sim 10^4$–10^7 cases, zero counterexamples) combinatorial lemma: the anti-correlated joint-finite recurrence, whose companion joint-infinite case—covering all unbounded-displacement permutations—is proven axiom-clean. The main theorem is therefore **machine-verified modulo one true, computationally-verified combinatorial lemma (the anti-correlated joint-finite recurrence); the joint-infinite case—covering all unbounded-displacement permutations—is proven axiom-clean**; it is not yet a complete, `sorry`-free proof.

Acknowledgements

The solution to this puzzle — including the characterisation of Bob's winning condition, the distinguishability theorem, the fixed-K impossibility argument, and the band-top lag-monotone staircase construction — was developed in collaboration with Claude (Anthropic). The simulation and belief-state game-solver code, the Lean 4 formalisation, and this writeup were subsequently produced in collaboration with Claude Code (Anthropic).